Paper numbers, package C

Pre-registration addendum C (`preregistration-addendum-C.md`) committed at ed05df2, before any episode of parts 1 to 4. Part 1 (E6) is confirmatory; parts 2 to 5 are exploratory or descriptive, with no test.

This research was supported by data provided by Meituan.

Summary

- E6 (confirmatory, n = 150): 141 of 144 contrasts established (Holm within E6); 0 of 72 cell-model pairs undo the cluster (none); 8 classes differ from those Paper A quotes.
- Stay (exploratory): on the 10 Q2 cells, W1 over 60 min falls below nearest courier's in 6 cells; Zhu et al. falls below `stay` in 8 and Jusup et al. in 8 (95% intervals below 0).
- Ashkrof setting at the first recalibration (exploratory): 0 of 128 classes change.
- Noisy demand (exploratory): 0 of 64 classes change.
- Ashkrof setting at the wide-grid recalibration (exploratory): 0 of 128 classes change.
- Ashkrof, first recalibration (its grid cut the beta0 interval): return probability 0.75 [0.65, 0.85] (fitted 0.75); stay constant -3 [-4, -2] (fitted -3).
- Ashkrof, wide-grid recalibration: return probability 0.75 [0.675, 0.85] (fitted 0.75); stay constant -4.5 [-6.5, -2.74] (fitted -3).
- Part 5, W1 by window after the cluster appears (0-60, 60-120, 180-240 min; model minus nearest courier, mean of the nine clustered cells): zhu -0.10, -0.16, -3.34; jusup -1.77, +0.76, -2.02; xu -0.48, +0.89, -0.70.
- Part 5, W1 over 20:00 to 20:59 by return probability (mean of the nine clustered cells): zhu p0 -0.21, fitted -3.34, p1 -4.19; jusup p0 +1.12, fitted -2.02, p1 -2.86; xu p0 +0.53, fitted -0.70, p1 +0.29.
- Part 5, Q1 W1 without 20:30 to 21:00: the largest shift is jusup, scattered, normal: +2.01 to +1.68.
- Part 5, per training policy: the largest spread between the five policy means of one contrast is 7.59 min (auad, E2, big_cluster, tight, p90_c2d_min).
- C6 (exploratory, homes at the drop-offs, seeds 1001 to 1030): cell-model pairs that undo the cluster: undo in 0 of 72 under `resid`, 0 under `indep`, 7 under the fitted setting.
- C7, fitted behavior, bands on a wider grid: return probability [0.675, 0.825] to [0.675, 0.825]; decision delay (s) [29.4, 42] to [29.4, 42]; cruise radius (m) [2000, 2800] to [1979, 2800] (point 2000 to 2800); stay constant (beta0) [-3.5, -2] to [-6, -2.5] (point -3 to -3.5). Point estimate moved for cruise radius (m), stay constant (beta0).
- Ashkrof variant, bands on a wider grid: return probability [0.65, 0.85] to [0.675, 0.85]; stay constant (beta0) [-4, -2] to [-6.5, -2.738] (point -3 to -4.5). Point estimate moved for stay constant (beta0).

Part 1, E6

Part 1. E6: clustered starts with preferred areas independent of the start (confirmatory)

Behavior `indep`, seeds 1001 to 1150 (n = 150), arm C at 100% obedience. 144 contrasts, Holm at 0.05 within E6; smallest effects as in E2. Differences are model minus nearest courier, in minutes; lower is better.

contrasts	established	established decrease of at least delta	established increase	equivalent within delta (R3)	inconclusive (R3)
144	141	8	93	3	0

141 established contrasts. Classes by model:

model	does not undo
auad	9
backpressure	9
foodmatch	9
forecast	9
jusup	9
li	9
xu	9
zhu	9

Cells whose class differs from the class Paper A quotes (E2, or E2-fresh for the frozen three): 8 of 72.

cell	model	W1 over 60 min [95% CI]	p_Holm	p90	p90 upper 95%	p90 p_Holm	E6 class	Paper A class (source)
mixed, tight	xu	-1.18 [-1.21, -1.14]	5.31e-114	+0.02	+0.44	1	does not undo	undoes at no delivery cost (E2)
mixed, normal	xu	-1.05 [-1.08, -1.03]	9.96e-123	+3.15	+3.18	1.12e-188	does not undo	undoes at no delivery cost (E2)
mixed, loose	xu	-0.90 [-0.92, -0.88]	6.79e-131	+3.20	+3.22	4.56e-196	does not undo	undoes at a delivery cost (E2)
three_clusters, normal	jusup	+1.79 [+1.74, +1.83]	1.43e-118	+1.95	+2.01	8.02e-107	does not undo	undoes at no delivery cost (E2)
three_clusters, loose	jusup	+1.40 [+1.37, +1.45]	2.3e-114	+0.57	+0.60	7.66e-90	does not undo	undoes at no delivery cost (E2)
mixed, tight	jusup	+2.97 [+2.91, +3.02]	3.65e-141	+9.24	+9.54	1.89e-102	does not undo	undoes at no delivery cost (E2)
mixed, normal	jusup	+2.51 [+2.46, +2.55]	1.53e-144	+1.20	+1.24	1.05e-100	does not undo	undoes at no delivery cost (E2)
mixed, loose	jusup	+1.89 [+1.85, +1.93]	3.21e-132	+0.27	+0.29	6.3e-58	does not undo	undoes at no delivery cost (E2)

Every cell: `c1_e6_classes.csv`; every contrast with its R3 reading: `c1_e6_contrasts.csv`.

Time to undo and the second-hour check (cells classed as undoing, and nearest courier)

Against the same model, obedience, supply level and seed from the scattered start (fitted rows of the full runs).

scenario	level	model	class	time_to_undo_median	time_to_undo_q25	time_to_undo_q75	censored_count	nearest median	second-hour W1	nearest second-hour W1	second_hour_
big_cluster	tight	nearest	nearest courier	55.000	54.000	58.000	0	55.000	5.660	5.660	
big_cluster	normal	nearest	nearest courier	91.000	88.000	96.000	0	91.000	7.510	7.510	
big_cluster	loose	nearest	nearest courier	132.000	120.000	146.750	1	132.000	8.070	8.070	
three_clusters	tight	nearest	nearest courier	65.000	63.000	67.000	0	65.000	5.350	5.350	
three_clusters	normal	nearest	nearest courier	132.500	119.000	145.000	2	132.500	8.850	8.850	
three_clusters	loose	nearest	nearest courier	159.000	133.250	192.500	18	159.000	7.200	7.200	
mixed	tight	nearest	nearest courier	59.000	56.000	63.000	0	59.000	5.570	5.570	
mixed	normal	nearest	nearest courier	66.000	63.000	69.000	0	66.000	4.570	4.570	
mixed	loose	nearest	nearest courier	72.000	68.000	77.000	0	72.000	4.160	4.160	

DQN (descriptive)

scenario	level	model	w1_60_minus_nearest	p90_minus_nearest
big_cluster	tight	jahanshahi	-0.015	-0.074
big_cluster	normal	jahanshahi	-0.036	0.007
big_cluster	loose	jahanshahi	-0.018	0.029
three_clusters	tight	jahanshahi	-0.001	-0.121
three_clusters	normal	jahanshahi	-0.018	0.011
three_clusters	loose	jahanshahi	-0.014	0.024
mixed	tight	jahanshahi	-0.064	0.004
mixed	normal	jahanshahi	-0.044	0.047
mixed	loose	jahanshahi	-0.031	0.032

Pre-registered secondary: W1 over 20:00 to 20:59, model minus nearest courier

model	big_cluster, loose	big_cluster, normal	big_cluster, tight	mixed, loose	mixed, normal	mixed, tight	three_clusters, loose	three_clusters, normal	three_clusters, tight
auad	+0.45 [+0.43, +0.48]	+0.00 [-0.02, +0.03]	-0.40 [-0.44, -0.37]	+0.15 [+0.13, +0.17]	-0.03 [-0.05, -0.00]	-0.08 [-0.12, -0.04]	+0.50 [+0.47, +0.52]	+0.39 [+0.36, +0.42]	-0.41 [-0.45, -0.38]
backpressure	-0.93 [-0.96, -0.91]	+0.23 [+0.20, +0.26]	-0.14 [-0.17, -0.11]	-0.19 [-0.21, -0.16]	-0.11 [-0.13, -0.08]	+0.15 [+0.11, +0.18]	-0.63 [-0.65, -0.60]	-0.45 [-0.48, -0.42]	-0.11 [-0.14, -0.07]
foodmatch	+0.35 [+0.33, +0.37]	+0.46 [+0.42, +0.50]	-0.62 [-0.66, -0.59]	+0.11 [+0.09, +0.14]	-0.06 [-0.08, -0.03]	-0.07 [-0.11, -0.03]	+0.43 [+0.40, +0.45]	+0.13 [+0.10, +0.16]	-0.66 [-0.70, -0.63]
forecast	+1.49 [+1.47, +1.52]	-0.25 [-0.29, -0.20]	-0.38 [-0.41, -0.35]	+0.52 [+0.48, +0.56]	-0.42 [-0.48, -0.37]	-0.23 [-0.26, -0.19]	+0.17 [+0.14, +0.20]	-0.10 [-0.13, -0.07]	-0.43 [-0.46, -0.40]
jahanshahi	-0.22 [-0.24, -0.20]	+0.03 [+0.01, +0.06]	-0.01 [-0.04, +0.02]	-0.08 [-0.10, -0.06]	-0.02 [-0.04, +0.00]	-0.03 [-0.05, -0.01]	-0.13 [-0.15, -0.11]	-0.18 [-0.20, -0.15]	-0.02 [-0.05, +0.01]
jusup	+4.96 [+4.90, +5.03]	+5.06 [+5.00, +5.11]	+0.80 [+0.76, +0.85]	+5.85 [+5.79, +5.91]	+4.14 [+4.08, +4.19]	+1.63 [+1.57, +1.69]	+8.30 [+8.24, +8.36]	+6.20 [+6.11, +6.29]	+0.97 [+0.92, +1.02]
li	-1.29 [-1.33, -1.25]	-0.08 [-0.11, -0.04]	-0.61 [-0.65, -0.56]	-0.56 [-0.59, -0.54]	-0.48 [-0.51, -0.45]	-0.33 [-0.36, -0.30]	-1.23 [-1.26, -1.19]	-0.88 [-0.91, -0.85]	-0.86 [-0.91, -0.82]
xu	+1.05 [+1.02, +1.09]	+0.45 [+0.41, +0.49]	-1.10 [-1.16, -1.04]	+0.68 [+0.65, +0.72]	+0.04 [-0.00, +0.08]	-0.79 [-0.85, -0.74]	+0.93 [+0.89, +0.97]	+0.81 [+0.76, +0.86]	-1.27 [-1.32, -1.21]
zhu	+1.06 [+1.01, +1.11]	+5.30 [+5.24, +5.36]	+0.43 [+0.39, +0.48]	+4.63 [+4.56, +4.71]	+2.22 [+2.12, +2.32]	+1.16 [+1.10, +1.22]	+7.56 [+7.49, +7.62]	+3.05 [+2.96, +3.14]	+0.55 [+0.50, +0.60]

Part 2, stay

Part 2. Stay without commands: suppression against repositioning (exploratory, no test)

Fitted behavior, seeds 1001 to 1150 (n = 150); `stay` and arm C at 100% obedience. Rows are cells, columns the contrasts; every number with its interval is in `c2_stay.csv`.

W1

cell	stay - nearest	zhu - stay	jusup - stay
scattered, tight	+1.05 [+1.02, +1.09]	-0.35 [-0.37, -0.33]	+0.11 [+0.09, +0.13]
scattered, normal	+1.61 [+1.59, +1.64]	-0.46 [-0.46, -0.45]	+0.39 [+0.37, +0.41]
scattered, loose	+1.24 [+1.22, +1.25]	-0.29 [-0.30, -0.29]	+0.32 [+0.30, +0.34]
burst, tight	+0.40 [+0.38, +0.41]	-0.20 [-0.21, -0.18]	+0.07 [+0.06, +0.09]
burst, normal	+1.71 [+1.69, +1.74]	-0.48 [-0.49, -0.47]	+0.33 [+0.32, +0.35]
burst, loose	+1.41 [+1.39, +1.43]	-0.32 [-0.33, -0.31]	+0.32 [+0.30, +0.34]
big_cluster, tight	+0.04 [+0.02, +0.05]	-0.09 [-0.10, -0.07]	+0.14 [+0.12, +0.16]

cell	stay - nearest	zhu - stay	jusup - stay
big_cluster, normal	-0.49 [-0.52, -0.45]	-0.33 [-0.34, -0.32]	+0.45 [+0.41, +0.48]
big_cluster, loose	-4.22 [-4.24, -4.19]	-0.23 [-0.24, -0.22]	+2.16 [+2.11, +2.20]
three_clusters, tight	+0.12 [+0.11, +0.14]	-0.08 [-0.09, -0.07]	-0.01 [-0.03, +0.01]
three_clusters, normal	-0.67 [-0.71, -0.63]	-0.12 [-0.14, -0.10]	+1.73 [+1.66, +1.80]
three_clusters, loose	-1.22 [-1.24, -1.20]	-0.26 [-0.27, -0.25]	-1.24 [-1.29, -1.19]
mixed, tight	+0.23 [+0.21, +0.24]	-0.10 [-0.12, -0.09]	-0.12 [-0.14, -0.10]
mixed, normal	-0.43 [-0.49, -0.36]	-0.28 [-0.29, -0.27]	+0.85 [+0.80, +0.90]
mixed, loose	-1.31 [-1.33, -1.29]	-0.18 [-0.19, -0.18]	-1.40 [-1.47, -1.33]
meetup, normal	+1.25 [+1.23, +1.27]	-0.41 [-0.42, -0.40]	+0.38 [+0.36, +0.40]

W1 over the first 60 min

cell	stay - nearest	zhu - stay	jusup - stay
scattered, tight	-0.52 [-0.53, -0.50]	-0.15 [-0.15, -0.14]	+0.28 [+0.26, +0.30]
scattered, normal	-0.78 [-0.79, -0.76]	-0.09 [-0.09, -0.09]	+0.28 [+0.26, +0.30]
scattered, loose	-1.06 [-1.07, -1.05]	-0.04 [-0.04, -0.04]	+0.33 [+0.31, +0.34]
burst, tight	-0.49 [-0.51, -0.48]	-0.15 [-0.16, -0.14]	+0.28 [+0.26, +0.30]
burst, normal	-0.73 [-0.74, -0.72]	-0.09 [-0.09, -0.08]	+0.28 [+0.26, +0.30]
burst, loose	-0.99 [-1.00, -0.98]	-0.04 [-0.04, -0.04]	+0.32 [+0.31, +0.34]
big_cluster, tight	-0.15 [-0.16, -0.14]	-0.01 [-0.01, -0.00]	+0.59 [+0.55, +0.63]
big_cluster, normal	-0.20 [-0.21, -0.19]	-0.02 [-0.02, -0.02]	-0.45 [-0.49, -0.41]
big_cluster, loose	-0.11 [-0.12, -0.10]	-0.02 [-0.02, -0.02]	-1.85 [-1.88, -1.81]
three_clusters, tight	-0.15 [-0.17, -0.14]	+0.04 [+0.03, +0.04]	-0.65 [-0.71, -0.59]
three_clusters, normal	-0.06 [-0.07, -0.04]	-0.09 [-0.10, -0.09]	-2.60 [-2.63, -2.56]
three_clusters, loose	+0.08 [+0.07, +0.10]	-0.14 [-0.15, -0.13]	-3.07 [-3.10, -3.04]
mixed, tight	-0.04 [-0.06, -0.03]	-0.00 [-0.01, +0.00]	-1.60 [-1.65, -1.54]
mixed, normal	+0.01 [+0.00, +0.02]	-0.02 [-0.03, -0.02]	-2.83 [-2.87, -2.79]
mixed, loose	+0.05 [+0.04, +0.06]	-0.03 [-0.03, -0.02]	-2.86 [-2.89, -2.83]
meetup, normal	+1.92 [+1.89, +1.95]	-0.53 [-0.55, -0.51]	+0.44 [+0.41, +0.47]

W1 over 20:00 to 20:59

cell	stay - nearest	zhu - stay	jusup - stay
scattered, tight	+3.20 [+3.14, +3.27]	-0.72 [-0.74, -0.69]	-0.16 [-0.19, -0.13]
scattered, normal	+3.80 [+3.76, +3.83]	-0.70 [-0.71, -0.69]	+0.28 [+0.26, +0.30]
scattered, loose	+3.17 [+3.14, +3.20]	-0.45 [-0.45, -0.44]	+0.24 [+0.20, +0.27]
burst, tight	+1.83 [+1.78, +1.88]	-0.52 [-0.55, -0.49]	-0.13 [-0.16, -0.10]
burst, normal	+3.95 [+3.91, +3.99]	-0.76 [-0.77, -0.75]	+0.05 [+0.03, +0.07]
burst, loose	+3.41 [+3.38, +3.44]	-0.51 [-0.51, -0.50]	+0.15 [+0.12, +0.18]
big_cluster, tight	+0.33 [+0.28, +0.39]	-0.34 [-0.37, -0.30]	+0.03 [-0.00, +0.07]
big_cluster, normal	-4.54 [-4.63, -4.46]	-0.65 [-0.66, -0.64]	-0.89 [-0.94, -0.84]
big_cluster, loose	-11.42 [-11.48, -11.37]	-0.43 [-0.44, -0.42]	+3.47 [+3.42, +3.51]
three_clusters, tight	+0.66 [+0.61, +0.70]	-0.37 [-0.41, -0.32]	+0.06 [+0.02, +0.10]
three_clusters, normal	-3.18 [-3.25, -3.11]	-0.49 [-0.51, -0.46]	+2.66 [+2.61, +2.72]
three_clusters, loose	-3.17 [-3.21, -3.12]	-0.34 [-0.35, -0.32]	+0.41 [+0.34, +0.47]
mixed, tight	+0.96 [+0.90, +1.01]	-0.43 [-0.47, -0.40]	+0.04 [+0.00, +0.07]
mixed, normal	-2.13 [-2.26, -1.99]	-0.61 [-0.63, -0.59]	+1.30 [+1.20, +1.41]
mixed, loose	-3.63 [-3.68, -3.57]	-0.32 [-0.34, -0.31]	+0.89 [+0.81, +0.98]
meetup, normal	+2.95 [+2.91, +2.99]	-0.64 [-0.66, -0.63]	+0.28 [+0.25, +0.31]

p90 order to door (min)

cell	stay - nearest	zhu - stay	jusup - stay
scattered, tight	-0.05 [-0.11, -0.00]	-0.32 [-0.38, -0.27]	+0.19 [+0.14, +0.24]
scattered, normal	+0.09 [+0.08, +0.09]	-0.09 [-0.09, -0.08]	-0.03 [-0.04, -0.03]
scattered, loose	+0.05 [+0.04, +0.05]	-0.05 [-0.06, -0.05]	-0.05 [-0.05, -0.04]
burst, tight	-0.08 [-0.14, -0.01]	-0.21 [-0.29, -0.14]	+0.25 [+0.17, +0.33]
burst, normal	+0.10 [+0.09, +0.10]	-0.11 [-0.12, -0.11]	-0.03 [-0.03, -0.02]
burst, loose	+0.06 [+0.05, +0.06]	-0.06 [-0.07, -0.06]	-0.05 [-0.05, -0.04]
big_cluster, tight	-0.01 [-0.12, +0.10]	-0.12 [-0.20, -0.04]	-1.64 [-1.75, -1.53]
big_cluster, normal	-0.06 [-0.09, -0.02]	-0.14 [-0.16, -0.12]	-2.20 [-2.28, -2.12]
big_cluster, loose	-0.02 [-0.04, -0.00]	-0.09 [-0.11, -0.08]	-3.04 [-3.08, -3.01]
three_clusters, tight	-0.26 [-0.37, -0.15]	-0.33 [-0.45, -0.22]	-2.84 [-2.97, -2.72]
three_clusters, normal	-0.25 [-0.29, -0.21]	-0.36 [-0.40, -0.33]	-3.11 [-3.16, -3.06]
three_clusters, loose	-0.11 [-0.13, -0.08]	-0.37 [-0.38, -0.35]	-2.17 [-2.21, -2.13]
mixed, tight	+0.03 [-0.06, +0.12]	-0.06 [-0.14, +0.03]	-2.21 [-2.35, -2.07]
mixed, normal	-0.03 [-0.05, -0.01]	-0.10 [-0.11, -0.08]	-1.50 [-1.53, -1.47]
mixed, loose	-0.02 [-0.03, -0.00]	-0.11 [-0.12, -0.10]	-0.65 [-0.68, -0.61]
meetup, normal	+0.09 [+0.08, +0.09]	-0.09 [-0.09, -0.08]	-0.03 [-0.04, -0.03]

km without an order per order

cell	stay - nearest	zhu - stay	jusup - stay
scattered, tight	-2.13 [-2.15, -2.11]	+0.49 [+0.47, +0.50]	+0.78 [+0.76, +0.80]
scattered, normal	-2.50 [-2.50, -2.49]	+0.66 [+0.65, +0.66]	+1.08 [+1.07, +1.10]
scattered, loose	-2.83 [-2.83, -2.82]	+0.70 [+0.70, +0.71]	+1.63 [+1.60, +1.65]
burst, tight	-2.29 [-2.30, -2.28]	+0.39 [+0.38, +0.40]	+0.73 [+0.71, +0.75]
burst, normal	-2.44 [-2.45, -2.44]	+0.63 [+0.62, +0.63]	+1.07 [+1.05, +1.09]
burst, loose	-2.79 [-2.79, -2.79]	+0.69 [+0.68, +0.69]	+1.57 [+1.55, +1.60]
big_cluster, tight	-1.90 [-1.91, -1.89]	+0.21 [+0.20, +0.22]	+1.03 [+1.00, +1.05]
big_cluster, normal	-3.08 [-3.09, -3.07]	+0.45 [+0.45, +0.46]	+1.49 [+1.47, +1.52]
big_cluster, loose	-3.36 [-3.37, -3.35]	+0.50 [+0.50, +0.51]	+2.42 [+2.40, +2.44]
three_clusters, tight	-2.06 [-2.08, -2.05]	+0.22 [+0.21, +0.23]	+0.82 [+0.80, +0.84]
three_clusters, normal	-2.73 [-2.74, -2.72]	+0.41 [+0.40, +0.41]	+1.36 [+1.34, +1.39]
three_clusters, loose	-2.58 [-2.59, -2.58]	+0.45 [+0.45, +0.46]	+2.80 [+2.76, +2.85]
mixed, tight	-2.09 [-2.10, -2.07]	+0.28 [+0.27, +0.29]	+0.92 [+0.90, +0.94]
mixed, normal	-2.67 [-2.68, -2.66]	+0.51 [+0.50, +0.51]	+1.57 [+1.55, +1.58]
mixed, loose	-2.54 [-2.54, -2.53]	+0.54 [+0.53, +0.54]	+3.08 [+3.05, +3.11]
meetup, normal	-2.45 [-2.46, -2.45]	+0.65 [+0.65, +0.66]	+1.08 [+1.07, +1.10]

Part 3, Ashkrof et al.

Part 3. One coherent behavior specification, Ashkrof et al. (exploratory, no test)

Recalibrated values beside the fitted behavior (95% bootstrap intervals over the Meituan moments):

parameter	fitted	fitted_ci_lo	fitted_ci_hi	ashkrof	ashkrof_ci_lo	ashkrof_ci_hi	ashkrof_note
p_return	0.750	0.675	0.825	0.750	0.650	0.850	recalibrated
asc_stay	-3.000	-3.500	-2.000	-3.000	-4.000	-2.000	recalibrated
epoch_s	42.000	29.400	42.000	42.000	42.000	42.000	fixed at the fitted value
cruise_radius_m	2000.000	2000.000	2800.000	2000.000	2000.000	2000.000	fixed at the fitted value
part_time_stay	nan	nan	nan	0.000	nan	nan	set to 0 (Scenario 1 Primary otherwise)
objective	2316.829	nan	nan	2591.285	nan	nan	

Moments (Meituan peak hours, the fitted behavior and the Ashkrof setting):

moment	in_fit	meituan_peak	meituan_all_day	simulated_fitted	simulated_ashkrof
exc_1000_15-30	False	0.431	0.443	0.494	0.493
exc_1000_5-15	False	0.335	0.345	0.129	0.131
exc_1000_<5	False	0.087	0.088	0.010	0.010
exc_1000_>=30	False	0.391	0.448	0.594	0.589
exc_1000_all	False	0.285	0.323	0.200	0.200
exc_2000_15-30	True	0.514	0.521	0.606	0.607
exc_2000_5-15	True	0.420	0.427	0.269	0.279
exc_2000_<5	True	0.098	0.105	0.049	0.048
exc_2000_>=30	True	0.490	0.539	0.695	0.698
exc_2000_all	False	0.369	0.406	0.308	0.310
exc_3000_15-30	False	0.501	0.522	0.587	0.590
exc_3000_5-15	False	0.399	0.414	0.242	0.246
exc_3000_<5	False	0.086	0.088	0.051	0.054
exc_3000_>=30	False	0.528	0.559	0.687	0.688
exc_3000_all	False	0.386	0.424	0.312	0.315
move_mean	nan	nan	nan	nan	1166.896
move_q10	True	43.393	73.282	0.000	0.000
move_q25	True	312.686	427.585	263.037	275.651
move_q50	True	910.804	1070.989	678.874	685.302
move_q75	True	1751.876	1961.133	1515.168	1527.966
move_q90	True	2748.672	2959.550	2864.637	2870.969
ret_1000_15-30	False	0.532	0.551	0.595	0.596
ret_1000_5-15	False	0.433	0.448	0.212	0.216
ret_1000_<5	False	0.130	0.131	0.035	0.035
ret_1000_>=30	False	0.484	0.555	0.687	0.683
ret_1000_all	False	0.366	0.413	0.263	0.264
ret_2000_15-30	True	0.690	0.705	0.788	0.788
ret_2000_5-15	True	0.600	0.611	0.402	0.414
ret_2000_<5	True	0.162	0.171	0.085	0.085
ret_2000_>=30	True	0.653	0.719	0.850	0.855
ret_2000_all	False	0.516	0.564	0.416	0.419
ret_3000_15-30	False	0.711	0.736	0.798	0.798
ret_3000_5-15	False	0.606	0.627	0.384	0.389
ret_3000_<5	False	0.158	0.160	0.093	0.097
ret_3000_>=30	False	0.728	0.768	0.876	0.877

moment	in_fit	meituan_peak	meituan_all_day	simulated_fitted	simulated_ashkrof
ret_3000_all	False	0.563	0.610	0.438	0.440
stay_15-30	True	0.035	0.040	0.038	0.034
stay_5-15	True	0.093	0.082	0.069	0.065
stay_<5	True	0.628	0.612	0.563	0.558
stay_>=30	True	0.030	0.031	0.041	0.033

Contrasts and classes

All ten models ran, seeds 1001 to 1030; the eight tested models are read against nearest courier, beside the fitted behavior's full-run rows at the same seeds, read the same way. Smallest effects per behavior: 10% of nearest courier's mean under that behavior (the fitted pilot for the fitted rows; seeds 1001 to 1030 of this part for `ashkrof`, which had no pilot). Holm within the E1-shaped and E2-shaped sets of each behavior, as a reading device. The DQN is a descriptive row (mean differences, no interval). Contrasts: `c3_ashkrof_contrasts.csv`; classes: `c3_ashkrof_classes.csv`; DQN: `c3_ashkrof_dqn.csv`.

W1 is W1 over the evening on Q1 cells and W1 over 60 min on Q2 cells; differences are model minus nearest courier, in minutes; lower is better.

input	holm_set	contrasts	established
ashkrof	E1-shaped	96	85
ashkrof	E2-shaped	160	134
fitted	E1-shaped	96	85
fitted	E2-shaped	160	132

Classes by model (cells):

model	level_1	ashkrof	fitted
auad	does not undo	10	10
auad	does not prevent	6	6
backpressure	does not undo	10	10
backpressure	does not prevent	4	4
backpressure	prevents at no delivery cost	2	2
foodmatch	does not undo	10	10
foodmatch	does not prevent	6	6
forecast	does not undo	10	10
forecast	prevents at no delivery cost	4	4
forecast	does not prevent	2	2
jusup	does not prevent	6	6
jusup	does not undo	6	6
jusup	undoes at no delivery cost	4	4
li	does not undo	10	10
li	prevents at no delivery cost	4	4
li	does not prevent	2	2
xu	does not undo	6	6
xu	does not prevent	4	4
xu	undoes at no delivery cost	2	2
xu	undoes at a delivery cost	2	2
xu	prevents at a delivery cost	1	1
xu	prevents at no delivery cost	1	1
zhu	does not undo	10	10
zhu	does not prevent	6	6

Classes that change: 0 of 128.

None.

Part 4, noisy demand

Part 4. Noisy demand information (exploratory, no test)

Forecast-aware matching, Li et al., Zhu et al. and Jusup et al. read the expected demand times a lognormal factor per zone and episode (mean 1, CV 0.2); W1 keeps the true demand. Seeds 1001 to 1030; nearest courier and the true-input rows come from the full runs at the same seeds. Holm runs within the E1-shaped and E2-shaped sets of each input, as a reading device. The noisy and true intervals of one contrast share their bootstrap resamples (common random numbers), so their difference is not blurred by resampling. Episodes equal to the true-input episode in every metric: forecast 0 of 480, li 480 of 480, zhu 0 of 480, jusup 6 of 480; a model with all its episodes equal reads no expected demand at run time. Contrasts: `c4_noise_contrasts.csv`; classes: `c4_noise_classes.csv`; identity: `c4_noise_identity.csv`.

W1 is W1 over the evening on Q1 cells and W1 over 60 min on Q2 cells; differences are model minus nearest courier, in minutes; lower is better.

input	holm_set	contrasts	established
noisy	E1-shaped	48	39
noisy	E2-shaped	80	69
true	E1-shaped	48	39
true	E2-shaped	80	70

Classes by model (cells):

model	level_1	noisy	true
forecast	does not undo	10	10
forecast	prevents at no delivery cost	4	4
forecast	does not prevent	2	2
jusup	does not prevent	6	6
jusup	does not undo	6	6
jusup	undoes at no delivery cost	4	4
li	does not undo	10	10
li	prevents at no delivery cost	4	4
li	does not prevent	2	2
zhu	does not undo	10	10
zhu	does not prevent	6	6

Classes that change: 0 of 64.

None.

Part 5, tables from the existing runs

Package C, part 5: tables from the existing runs (descriptive)

Seeds 1001 to 1150 (n = 150), fitted behavior unless stated, arm C at 100% obedience. Differences are model minus nearest courier, paired by seed, with 95% percentile bootstrap intervals; no test.

W1 by window after the cluster appears

cell	window (min)	model	difference [95% CI]
big_cluster, tight	0-60	foodmatch	+0.50 [+0.49, +0.52]
big_cluster, tight	0-60	forecast	+0.01 [+0.01, +0.01]
big_cluster, tight	0-60	backpressure	-0.06 [-0.06, -0.06]
big_cluster, tight	0-60	xu	+3.34 [+3.31, +3.38]
big_cluster, tight	0-60	li	-0.07 [-0.07, -0.06]
big_cluster, tight	0-60	jahanshahi	+0.00 [-0.00, +0.00]
big_cluster, tight	0-60	auad	+0.30 [+0.22, +0.37]
big_cluster, tight	0-60	zhu	-0.16 [-0.17, -0.15]
big_cluster, tight	0-60	jusup	+0.44 [+0.39, +0.48]
big_cluster, tight	60-120	foodmatch	-0.16 [-0.20, -0.13]
big_cluster, tight	60-120	forecast	+0.01 [-0.00, +0.02]
big_cluster, tight	60-120	backpressure	-0.00 [-0.02, +0.01]
big_cluster, tight	60-120	xu	+1.26 [+1.22, +1.30]
big_cluster, tight	60-120	li	+0.19 [+0.17, +0.22]
big_cluster, tight	60-120	jahanshahi	+0.01 [-0.00, +0.02]
big_cluster, tight	60-120	auad	-0.01 [-0.03, +0.02]
big_cluster, tight	60-120	zhu	-0.01 [-0.03, +0.01]
big_cluster, tight	60-120	jusup	-0.14 [-0.16, -0.12]
big_cluster, tight	180-240	foodmatch	+4.14 [+4.07, +4.21]
big_cluster, tight	180-240	forecast	-0.04 [-0.07, -0.00]
big_cluster, tight	180-240	backpressure	-0.23 [-0.27, -0.19]
big_cluster, tight	180-240	xu	+1.75 [+1.68, +1.82]
big_cluster, tight	180-240	li	+4.17 [+4.10, +4.26]
big_cluster, tight	180-240	jahanshahi	-0.03 [-0.07, +0.01]
big_cluster, tight	180-240	auad	+1.34 [+1.21, +1.46]
big_cluster, tight	180-240	zhu	-0.00 [-0.07, +0.06]
big_cluster, tight	180-240	jusup	+0.37 [+0.31, +0.43]
big_cluster, normal	0-60	foodmatch	+0.17 [+0.16, +0.18]
big_cluster, normal	0-60	forecast	+0.01 [+0.01, +0.02]
big_cluster, normal	0-60	backpressure	-0.10 [-0.10, -0.10]
big_cluster, normal	0-60	xu	+0.90 [+0.88, +0.93]
big_cluster, normal	0-60	li	-0.12 [-0.13, -0.12]
big_cluster, normal	0-60	jahanshahi	-0.00 [-0.00, -0.00]
big_cluster, normal	0-60	auad	+0.11 [+0.08, +0.13]
big_cluster, normal	0-60	zhu	-0.22 [-0.23, -0.21]
big_cluster, normal	0-60	jusup	-0.65 [-0.69, -0.61]
big_cluster, normal	60-120	foodmatch	+0.80 [+0.73, +0.88]
big_cluster, normal	60-120	forecast	-0.25 [-0.26, -0.23]
big_cluster, normal	60-120	backpressure	-0.38 [-0.40, -0.36]
big_cluster, normal	60-120	xu	+3.36 [+3.28, +3.45]
big_cluster, normal	60-120	li	-0.28 [-0.32, -0.24]
big_cluster, normal	60-120	jahanshahi	-0.00 [-0.02, +0.01]
big_cluster, normal	60-120	auad	-0.01 [-0.06, +0.05]

cell	window (min)	model	difference [95% CI]
big_cluster, normal	60-120	zhu	+0.30 [+0.27, +0.33]
big_cluster, normal	60-120	jusup	+4.76 [+4.62, +4.89]
big_cluster, normal	180-240	foodmatch	-0.10 [-0.13, -0.06]
big_cluster, normal	180-240	forecast	+0.40 [+0.37, +0.42]
big_cluster, normal	180-240	backpressure	-0.96 [-0.99, -0.93]
big_cluster, normal	180-240	xu	-4.55 [-4.61, -4.50]
big_cluster, normal	180-240	li	-1.24 [-1.27, -1.20]
big_cluster, normal	180-240	jahanshahi	-0.06 [-0.08, -0.03]
big_cluster, normal	180-240	auad	-0.04 [-0.07, -0.01]
big_cluster, normal	180-240	zhu	-5.19 [-5.28, -5.10]
big_cluster, normal	180-240	jusup	-5.44 [-5.53, -5.34]
big_cluster, loose	0-60	foodmatch	+0.10 [+0.10, +0.11]
big_cluster, loose	0-60	forecast	+0.02 [+0.02, +0.02]
big_cluster, loose	0-60	backpressure	-0.06 [-0.06, -0.05]
big_cluster, loose	0-60	xu	+0.25 [+0.24, +0.25]
big_cluster, loose	0-60	li	-0.08 [-0.09, -0.08]
big_cluster, loose	0-60	jahanshahi	-0.00 [-0.00, -0.00]
big_cluster, loose	0-60	auad	+0.04 [+0.04, +0.05]
big_cluster, loose	0-60	zhu	-0.13 [-0.14, -0.12]
big_cluster, loose	0-60	jusup	-1.96 [-1.99, -1.93]
big_cluster, loose	60-120	foodmatch	-0.79 [-0.83, -0.75]
big_cluster, loose	60-120	forecast	+0.27 [+0.26, +0.28]
big_cluster, loose	60-120	backpressure	-0.41 [-0.42, -0.39]
big_cluster, loose	60-120	xu	+3.66 [+3.61, +3.72]
big_cluster, loose	60-120	li	-0.57 [-0.59, -0.55]
big_cluster, loose	60-120	jahanshahi	-0.06 [-0.07, -0.05]
big_cluster, loose	60-120	auad	+0.56 [+0.48, +0.65]
big_cluster, loose	60-120	zhu	-0.72 [-0.74, -0.70]
big_cluster, loose	60-120	jusup	+2.41 [+2.32, +2.50]
big_cluster, loose	180-240	foodmatch	-0.63 [-0.66, -0.61]
big_cluster, loose	180-240	forecast	+0.36 [+0.34, +0.38]
big_cluster, loose	180-240	backpressure	-0.91 [-0.93, -0.89]
big_cluster, loose	180-240	xu	-3.87 [-3.92, -3.83]
big_cluster, loose	180-240	li	-1.31 [-1.34, -1.29]
big_cluster, loose	180-240	jahanshahi	-0.06 [-0.08, -0.05]
big_cluster, loose	180-240	auad	+0.00 [-0.02, +0.03]
big_cluster, loose	180-240	zhu	-11.86 [-11.90, -11.81]
big_cluster, loose	180-240	jusup	-7.96 [-8.02, -7.89]
three_clusters, tight	0-60	foodmatch	-1.20 [-1.23, -1.18]
three_clusters, tight	0-60	forecast	-0.04 [-0.05, -0.04]
three_clusters, tight	0-60	backpressure	-0.06 [-0.06, -0.05]
three_clusters, tight	0-60	xu	-0.07 [-0.14, +0.01]
three_clusters, tight	0-60	li	-0.03 [-0.04, -0.02]
three_clusters, tight	0-60	jahanshahi	-0.00 [-0.00, +0.00]
three_clusters, tight	0-60	auad	-0.08 [-0.11, -0.05]
three_clusters, tight	0-60	zhu	-0.12 [-0.13, -0.10]
three_clusters, tight	0-60	jusup	-0.80 [-0.86, -0.74]
three_clusters, tight	60-120	foodmatch	-0.24 [-0.27, -0.22]
three_clusters, tight	60-120	forecast	-0.00 [-0.01, +0.01]
three_clusters, tight	60-120	backpressure	+0.00 [-0.01, +0.02]
three_clusters, tight	60-120	xu	+3.67 [+3.58, +3.75]
three_clusters, tight	60-120	li	+0.33 [+0.30, +0.35]
three_clusters, tight	60-120	jahanshahi	-0.00 [-0.01, +0.01]
three_clusters, tight	60-120	auad	+0.01 [-0.01, +0.04]
three_clusters, tight	60-120	zhu	+0.01 [-0.01, +0.03]
three_clusters, tight	60-120	jusup	+0.54 [+0.48, +0.61]
three_clusters, tight	180-240	foodmatch	+3.30 [+3.22, +3.37]
three_clusters, tight	180-240	forecast	-0.21 [-0.24, -0.17]
three_clusters, tight	180-240	backpressure	-0.14 [-0.17, -0.11]
three_clusters, tight	180-240	xu	+0.54 [+0.47, +0.61]
three_clusters, tight	180-240	li	+1.04 [+0.97, +1.11]
three_clusters, tight	180-240	jahanshahi	+0.02 [-0.02, +0.05]
three_clusters, tight	180-240	auad	+0.72 [+0.62, +0.84]
three_clusters, tight	180-240	zhu	+0.29 [+0.25, +0.34]
three_clusters, tight	180-240	jusup	+0.72 [+0.67, +0.76]
three_clusters, normal	0-60	foodmatch	-1.03 [-1.05, -1.01]
three_clusters, normal	0-60	forecast	-0.02 [-0.03, -0.02]
three_clusters, normal	0-60	backpressure	-0.10 [-0.10, -0.09]

cell	window (min)	model	difference [95% CI]
three_clusters, normal	0-60	xu	-1.45 [-1.48, -1.42]
three_clusters, normal	0-60	li	-0.22 [-0.23, -0.21]
three_clusters, normal	0-60	jahanshahi	-0.01 [-0.01, -0.01]
three_clusters, normal	0-60	auad	-0.18 [-0.19, -0.17]
three_clusters, normal	0-60	zhu	-0.15 [-0.17, -0.14]
three_clusters, normal	0-60	jusup	-2.65 [-2.69, -2.62]
three_clusters, normal	60-120	foodmatch	-1.98 [-2.07, -1.89]
three_clusters, normal	60-120	forecast	-0.18 [-0.20, -0.15]
three_clusters, normal	60-120	backpressure	+0.01 [-0.03, +0.04]
three_clusters, normal	60-120	xu	+0.21 [+0.02, +0.42]
three_clusters, normal	60-120	li	+0.62 [+0.54, +0.70]
three_clusters, normal	60-120	jahanshahi	+0.02 [+0.00, +0.05]
three_clusters, normal	60-120	auad	-0.09 [-0.13, -0.04]
three_clusters, normal	60-120	zhu	+0.35 [+0.30, +0.40]
three_clusters, normal	60-120	jusup	+3.12 [+2.92, +3.32]
three_clusters, normal	180-240	foodmatch	+1.89 [+1.83, +1.95]
three_clusters, normal	180-240	forecast	-0.11 [-0.15, -0.08]
three_clusters, normal	180-240	backpressure	-0.31 [-0.35, -0.26]
three_clusters, normal	180-240	xu	+0.18 [+0.11, +0.26]
three_clusters, normal	180-240	li	-0.19 [-0.25, -0.14]
three_clusters, normal	180-240	jahanshahi	-0.03 [-0.06, -0.00]
three_clusters, normal	180-240	auad	+0.60 [+0.54, +0.66]
three_clusters, normal	180-240	zhu	-3.67 [-3.74, -3.60]
three_clusters, normal	180-240	jusup	-0.52 [-0.59, -0.44]
three_clusters, loose	0-60	foodmatch	-0.81 [-0.83, -0.80]
three_clusters, loose	0-60	forecast	-0.02 [-0.02, -0.01]
three_clusters, loose	0-60	backpressure	-0.10 [-0.11, -0.09]
three_clusters, loose	0-60	xu	-1.11 [-1.13, -1.08]
three_clusters, loose	0-60	li	-0.22 [-0.23, -0.21]
three_clusters, loose	0-60	jahanshahi	-0.01 [-0.01, -0.01]
three_clusters, loose	0-60	auad	-0.16 [-0.17, -0.15]
three_clusters, loose	0-60	zhu	-0.06 [-0.07, -0.05]
three_clusters, loose	0-60	jusup	-2.98 [-3.02, -2.95]
three_clusters, loose	60-120	foodmatch	-1.82 [-1.86, -1.79]
three_clusters, loose	60-120	forecast	-0.03 [-0.05, -0.02]
three_clusters, loose	60-120	backpressure	-0.22 [-0.24, -0.21]
three_clusters, loose	60-120	xu	-2.26 [-2.32, -2.20]
three_clusters, loose	60-120	li	-0.77 [-0.80, -0.73]
three_clusters, loose	60-120	jahanshahi	-0.06 [-0.07, -0.05]
three_clusters, loose	60-120	auad	-0.20 [-0.22, -0.17]
three_clusters, loose	60-120	zhu	-0.78 [-0.81, -0.76]
three_clusters, loose	60-120	jusup	-2.98 [-3.04, -2.91]
three_clusters, loose	180-240	foodmatch	-0.06 [-0.09, -0.04]
three_clusters, loose	180-240	forecast	-0.04 [-0.06, -0.02]
three_clusters, loose	180-240	backpressure	-0.36 [-0.38, -0.34]
three_clusters, loose	180-240	xu	-1.63 [-1.67, -1.60]
three_clusters, loose	180-240	li	-0.54 [-0.56, -0.51]
three_clusters, loose	180-240	jahanshahi	-0.04 [-0.06, -0.03]
three_clusters, loose	180-240	auad	+0.13 [+0.11, +0.16]
three_clusters, loose	180-240	zhu	-3.50 [-3.55, -3.46]
three_clusters, loose	180-240	jusup	-2.76 [-2.81, -2.71]
mixed, tight	0-60	foodmatch	-0.08 [-0.08, -0.07]
mixed, tight	0-60	forecast	+0.08 [+0.08, +0.09]
mixed, tight	0-60	backpressure	-0.08 [-0.09, -0.07]
mixed, tight	0-60	xu	-2.08 [-2.12, -2.05]
mixed, tight	0-60	li	-0.40 [-0.41, -0.39]
mixed, tight	0-60	jahanshahi	-0.02 [-0.02, -0.02]
mixed, tight	0-60	auad	+0.05 [+0.04, +0.06]
mixed, tight	0-60	zhu	-0.05 [-0.06, -0.03]
mixed, tight	0-60	jusup	-1.64 [-1.70, -1.59]
mixed, tight	60-120	foodmatch	-0.37 [-0.40, -0.34]
mixed, tight	60-120	forecast	-0.02 [-0.04, -0.01]
mixed, tight	60-120	backpressure	-0.02 [-0.03, -0.00]
mixed, tight	60-120	xu	+0.21 [+0.16, +0.26]
mixed, tight	60-120	li	+0.10 [+0.08, +0.12]
mixed, tight	60-120	jahanshahi	-0.01 [-0.02, +0.01]
mixed, tight	60-120	auad	+0.00 [-0.02, +0.02]
mixed, tight	60-120	zhu	-0.01 [-0.03, +0.00]

cell	window (min)	model	difference [95% CI]
mixed, tight	60-120	jusup	+0.91 [+0.86, +0.95]
mixed, tight	180-240	foodmatch	+1.99 [+1.93, +2.05]
mixed, tight	180-240	forecast	-0.09 [-0.12, -0.06]
mixed, tight	180-240	backpressure	-0.23 [-0.26, -0.20]
mixed, tight	180-240	xu	+0.92 [+0.86, +0.97]
mixed, tight	180-240	li	+0.73 [+0.67, +0.79]
mixed, tight	180-240	jahanshahi	-0.04 [-0.07, -0.00]
mixed, tight	180-240	auad	+0.11 [+0.05, +0.17]
mixed, tight	180-240	zhu	+0.52 [+0.47, +0.58]
mixed, tight	180-240	jusup	+1.00 [+0.94, +1.06]
mixed, normal	0-60	foodmatch	+0.08 [+0.08, +0.09]
mixed, normal	0-60	forecast	+0.18 [+0.18, +0.19]
mixed, normal	0-60	backpressure	-0.10 [-0.10, -0.09]
mixed, normal	0-60	xu	-2.38 [-2.41, -2.36]
mixed, normal	0-60	li	-0.55 [-0.56, -0.54]
mixed, normal	0-60	jahanshahi	-0.03 [-0.03, -0.03]
mixed, normal	0-60	auad	+0.08 [+0.08, +0.08]
mixed, normal	0-60	zhu	-0.01 [-0.02, +0.00]
mixed, normal	0-60	jusup	-2.82 [-2.86, -2.78]
mixed, normal	60-120	foodmatch	-0.23 [-0.27, -0.20]
mixed, normal	60-120	forecast	-0.09 [-0.11, -0.08]
mixed, normal	60-120	backpressure	-0.17 [-0.20, -0.14]
mixed, normal	60-120	xu	+0.33 [+0.24, +0.42]
mixed, normal	60-120	li	-0.25 [-0.29, -0.21]
mixed, normal	60-120	jahanshahi	-0.01 [-0.02, +0.01]
mixed, normal	60-120	auad	-0.13 [-0.16, -0.10]
mixed, normal	60-120	zhu	-0.16 [-0.19, -0.14]
mixed, normal	60-120	jusup	+1.93 [+1.80, +2.07]
mixed, normal	180-240	foodmatch	+0.69 [+0.66, +0.73]
mixed, normal	180-240	forecast	+0.69 [+0.65, +0.72]
mixed, normal	180-240	backpressure	-0.69 [-0.72, -0.66]
mixed, normal	180-240	xu	+0.68 [+0.64, +0.73]
mixed, normal	180-240	li	-1.02 [-1.06, -0.98]
mixed, normal	180-240	jahanshahi	-0.05 [-0.08, -0.02]
mixed, normal	180-240	auad	-0.11 [-0.15, -0.07]
mixed, normal	180-240	zhu	-2.74 [-2.87, -2.61]
mixed, normal	180-240	jusup	-0.82 [-0.90, -0.75]
mixed, loose	0-60	foodmatch	+0.06 [+0.05, +0.06]
mixed, loose	0-60	forecast	+0.26 [+0.26, +0.27]
mixed, loose	0-60	backpressure	-0.12 [-0.12, -0.11]
mixed, loose	0-60	xu	-1.77 [-1.79, -1.75]
mixed, loose	0-60	li	-0.50 [-0.51, -0.49]
mixed, loose	0-60	jahanshahi	-0.03 [-0.03, -0.03]
mixed, loose	0-60	auad	+0.07 [+0.06, +0.07]
mixed, loose	0-60	zhu	+0.02 [+0.01, +0.03]
mixed, loose	0-60	jusup	-2.81 [-2.84, -2.78]
mixed, loose	60-120	foodmatch	+0.14 [+0.13, +0.16]
mixed, loose	60-120	forecast	+0.36 [+0.35, +0.38]
mixed, loose	60-120	backpressure	-0.24 [-0.25, -0.22]
mixed, loose	60-120	xu	-2.40 [-2.46, -2.35]
mixed, loose	60-120	li	-1.82 [-1.85, -1.79]
mixed, loose	60-120	jahanshahi	-0.13 [-0.14, -0.12]
mixed, loose	60-120	auad	+0.25 [+0.23, +0.26]
mixed, loose	60-120	zhu	-0.39 [-0.40, -0.37]
mixed, loose	60-120	jusup	-3.69 [-3.77, -3.60]
mixed, loose	180-240	foodmatch	+0.13 [+0.10, +0.15]
mixed, loose	180-240	forecast	+0.57 [+0.55, +0.59]
mixed, loose	180-240	backpressure	-0.38 [-0.40, -0.36]
mixed, loose	180-240	xu	-0.31 [-0.35, -0.27]
mixed, loose	180-240	li	-1.16 [-1.19, -1.13]
mixed, loose	180-240	jahanshahi	-0.04 [-0.06, -0.03]
mixed, loose	180-240	auad	-0.06 [-0.08, -0.04]
mixed, loose	180-240	zhu	-3.95 [-4.00, -3.90]
mixed, loose	180-240	jusup	-2.74 [-2.80, -2.68]
meetup, normal	0-60	foodmatch	+0.05 [+0.02, +0.07]
meetup, normal	0-60	forecast	-0.43 [-0.47, -0.40]
meetup, normal	0-60	backpressure	-0.14 [-0.17, -0.11]
meetup, normal	0-60	xu	-2.99 [-3.16, -2.82]

cell	window (min)	model	difference [95% CI]
meetup, normal	0-60	li	-0.49 [-0.53, -0.46]
meetup, normal	0-60	jahanshahi	-0.04 [-0.06, -0.02]
meetup, normal	0-60	auad	+0.00 [-0.02, +0.03]
meetup, normal	0-60	zhu	+1.39 [+1.36, +1.43]
meetup, normal	0-60	jusup	+2.36 [+2.31, +2.40]
meetup, normal	60-120	foodmatch	+0.08 [+0.06, +0.11]
meetup, normal	60-120	forecast	-0.34 [-0.37, -0.31]
meetup, normal	60-120	backpressure	-0.26 [-0.29, -0.23]
meetup, normal	60-120	xu	-3.38 [-3.44, -3.32]
meetup, normal	60-120	li	-0.68 [-0.71, -0.65]
meetup, normal	60-120	jahanshahi	-0.07 [-0.09, -0.04]
meetup, normal	60-120	auad	+0.03 [+0.01, +0.06]
meetup, normal	60-120	zhu	+2.31 [+2.27, +2.35]
meetup, normal	60-120	jusup	+3.23 [+3.18, +3.28]

Q1 W1 without 20:30 to 21:00

cell	model	full evening	without 20:30-21:00
scattered, tight	foodmatch	+0.06 [+0.05, +0.07]	+0.02 [+0.01, +0.04]
scattered, tight	forecast	-0.41 [-0.43, -0.40]	-0.39 [-0.41, -0.38]
scattered, tight	backpressure	-0.20 [-0.21, -0.18]	-0.22 [-0.23, -0.20]
scattered, tight	xu	-0.68 [-0.72, -0.64]	-0.69 [-0.73, -0.65]
scattered, tight	li	-0.31 [-0.33, -0.29]	-0.31 [-0.33, -0.29]
scattered, tight	jahanshahi	-0.03 [-0.04, -0.02]	-0.03 [-0.04, -0.02]
scattered, tight	auad	-0.07 [-0.09, -0.06]	-0.10 [-0.11, -0.08]
scattered, tight	zhu	+0.70 [+0.67, +0.72]	+0.39 [+0.37, +0.42]
scattered, tight	jusup	+1.16 [+1.11, +1.21]	+0.84 [+0.80, +0.89]
scattered, normal	foodmatch	+0.09 [+0.08, +0.10]	+0.07 [+0.06, +0.08]
scattered, normal	forecast	-0.69 [-0.71, -0.67]	-0.69 [-0.70, -0.67]
scattered, normal	backpressure	-0.29 [-0.31, -0.28]	-0.28 [-0.30, -0.27]
scattered, normal	xu	+0.22 [+0.18, +0.26]	+0.29 [+0.25, +0.34]
scattered, normal	li	-0.68 [-0.70, -0.66]	-0.66 [-0.68, -0.64]
scattered, normal	jahanshahi	-0.07 [-0.08, -0.06]	-0.06 [-0.07, -0.05]
scattered, normal	auad	+0.09 [+0.08, +0.10]	+0.07 [+0.06, +0.08]
scattered, normal	zhu	+1.16 [+1.13, +1.18]	+0.85 [+0.83, +0.87]
scattered, normal	jusup	+2.01 [+1.98, +2.04]	+1.68 [+1.65, +1.71]
scattered, loose	foodmatch	+0.09 [+0.08, +0.10]	+0.08 [+0.07, +0.09]
scattered, loose	forecast	-0.57 [-0.58, -0.55]	-0.57 [-0.58, -0.55]
scattered, loose	backpressure	-0.46 [-0.47, -0.44]	-0.44 [-0.45, -0.43]
scattered, loose	xu	+0.67 [+0.64, +0.71]	+0.76 [+0.72, +0.79]
scattered, loose	li	-0.79 [-0.81, -0.78]	-0.76 [-0.77, -0.74]
scattered, loose	jahanshahi	-0.09 [-0.09, -0.08]	-0.08 [-0.09, -0.07]
scattered, loose	auad	+0.10 [+0.10, +0.11]	+0.09 [+0.09, +0.10]
scattered, loose	zhu	+0.95 [+0.93, +0.96]	+0.67 [+0.65, +0.69]
scattered, loose	jusup	+1.55 [+1.53, +1.58]	+1.27 [+1.24, +1.30]
burst, tight	foodmatch	-0.14 [-0.15, -0.12]	-0.13 [-0.15, -0.11]
burst, tight	forecast	-0.23 [-0.24, -0.21]	-0.19 [-0.20, -0.18]
burst, tight	backpressure	-0.15 [-0.17, -0.14]	-0.18 [-0.19, -0.17]
burst, tight	xu	-0.72 [-0.75, -0.70]	-0.69 [-0.71, -0.66]
burst, tight	li	-0.39 [-0.41, -0.37]	-0.35 [-0.38, -0.33]
burst, tight	jahanshahi	-0.01 [-0.02, +0.00]	-0.01 [-0.02, -0.00]
burst, tight	auad	-0.12 [-0.13, -0.11]	-0.10 [-0.11, -0.09]
burst, tight	zhu	+0.20 [+0.18, +0.22]	-0.04 [-0.05, -0.02]
burst, tight	jusup	+0.47 [+0.45, +0.49]	+0.21 [+0.20, +0.23]
burst, normal	foodmatch	+0.06 [+0.05, +0.07]	+0.04 [+0.03, +0.05]
burst, normal	forecast	-0.57 [-0.58, -0.56]	-0.56 [-0.58, -0.55]
burst, normal	backpressure	-0.27 [-0.29, -0.26]	-0.26 [-0.28, -0.25]
burst, normal	xu	-0.22 [-0.26, -0.18]	-0.20 [-0.24, -0.16]
burst, normal	li	-0.63 [-0.65, -0.61]	-0.61 [-0.62, -0.59]
burst, normal	jahanshahi	-0.07 [-0.08, -0.06]	-0.06 [-0.07, -0.05]
burst, normal	auad	+0.07 [+0.06, +0.09]	+0.05 [+0.04, +0.07]
burst, normal	zhu	+1.23 [+1.21, +1.26]	+0.93 [+0.90, +0.95]
burst, normal	jusup	+2.05 [+2.02, +2.08]	+1.75 [+1.72, +1.78]
burst, loose	foodmatch	+0.08 [+0.07, +0.09]	+0.07 [+0.06, +0.08]
burst, loose	forecast	-0.54 [-0.55, -0.53]	-0.53 [-0.55, -0.52]
burst, loose	backpressure	-0.42 [-0.43, -0.41]	-0.40 [-0.42, -0.39]
burst, loose	xu	+0.44 [+0.41, +0.47]	+0.50 [+0.46, +0.53]
burst, loose	li	-0.76 [-0.78, -0.75]	-0.72 [-0.74, -0.71]

cell	model	full evening	without 20:30-21:00
burst, loose	jahanshahi	-0.07 [-0.08, -0.06]	-0.06 [-0.07, -0.06]
burst, loose	auad	+0.09 [+0.08, +0.10]	+0.08 [+0.07, +0.09]
burst, loose	zhu	+1.09 [+1.07, +1.11]	+0.81 [+0.80, +0.83]
burst, loose	jusup	+1.73 [+1.70, +1.76]	+1.45 [+1.43, +1.48]

Per training policy: range of the five policy means of each E1 and E2 contrast

Every policy mean with its interval: `c5_per_policy.csv`.

family	model	scenario	level	metric	min	max
E1	auad	burst	loose	p90_c2d_min	0.347	0.370
E1	auad	burst	loose	w1_rule_min	0.080	0.118
E1	auad	burst	normal	p90_c2d_min	0.424	0.449
E1	auad	burst	normal	w1_rule_min	0.055	0.100
E1	auad	burst	tight	p90_c2d_min	-1.651	1.025
E1	auad	burst	tight	w1_rule_min	-0.161	-0.100
E1	auad	scattered	loose	p90_c2d_min	0.332	0.350
E1	auad	scattered	loose	w1_rule_min	0.094	0.112
E1	auad	scattered	normal	p90_c2d_min	0.401	0.437
E1	auad	scattered	normal	w1_rule_min	0.074	0.113
E1	auad	scattered	tight	p90_c2d_min	0.506	0.835
E1	auad	scattered	tight	w1_rule_min	-0.084	-0.051
E1	jusup	burst	loose	p90_c2d_min	0.002	0.020
E1	jusup	burst	loose	w1_rule_min	1.585	1.834
E1	jusup	burst	normal	p90_c2d_min	0.042	0.095
E1	jusup	burst	normal	w1_rule_min	1.962	2.169
E1	jusup	burst	tight	p90_c2d_min	0.058	0.329
E1	jusup	burst	tight	w1_rule_min	0.448	0.496
E1	jusup	scattered	loose	p90_c2d_min	-0.010	0.008
E1	jusup	scattered	loose	w1_rule_min	1.367	1.665
E1	jusup	scattered	normal	p90_c2d_min	0.031	0.062
E1	jusup	scattered	normal	w1_rule_min	1.891	2.171
E1	jusup	scattered	tight	p90_c2d_min	0.039	0.246
E1	jusup	scattered	tight	w1_rule_min	1.106	1.218
E1	li	burst	loose	p90_c2d_min	0.499	0.533
E1	li	burst	loose	w1_rule_min	-0.801	-0.712
E1	li	burst	normal	p90_c2d_min	0.379	0.412
E1	li	burst	normal	w1_rule_min	-0.692	-0.575
E1	li	burst	tight	p90_c2d_min	-5.118	-4.508
E1	li	burst	tight	w1_rule_min	-0.436	-0.365
E1	li	scattered	loose	p90_c2d_min	0.515	0.552
E1	li	scattered	loose	w1_rule_min	-0.860	-0.746
E1	li	scattered	normal	p90_c2d_min	0.452	0.479
E1	li	scattered	normal	w1_rule_min	-0.750	-0.624
E1	li	scattered	tight	p90_c2d_min	-0.316	-0.029
E1	li	scattered	tight	w1_rule_min	-0.336	-0.274
E1	xu	burst	loose	p90_c2d_min	3.339	3.411
E1	xu	burst	loose	w1_rule_min	0.395	0.481
E1	xu	burst	normal	p90_c2d_min	3.245	3.312
E1	xu	burst	normal	w1_rule_min	-0.280	-0.138
E1	xu	burst	tight	p90_c2d_min	-4.929	-4.347
E1	xu	burst	tight	w1_rule_min	-0.751	-0.685
E1	xu	scattered	loose	p90_c2d_min	3.186	3.242
E1	xu	scattered	loose	w1_rule_min	0.622	0.745
E1	xu	scattered	normal	p90_c2d_min	3.130	3.212
E1	xu	scattered	normal	w1_rule_min	0.140	0.271
E1	xu	scattered	tight	p90_c2d_min	1.710	2.171
E1	xu	scattered	tight	w1_rule_min	-0.741	-0.640
E2	auad	big_cluster	loose	p90_c2d_min	0.566	1.302
E2	auad	big_cluster	loose	w1_rule_60min_after_min	0.024	0.091
E2	auad	big_cluster	normal	p90_c2d_min	0.771	1.131
E2	auad	big_cluster	normal	w1_rule_60min_after_min	0.015	0.393
E2	auad	big_cluster	tight	p90_c2d_min	-6.229	1.363
E2	auad	big_cluster	tight	w1_rule_60min_after_min	0.002	1.237
E2	auad	meetup	normal	p90_c2d_min	0.417	0.456
E2	auad	meetup	normal	w1_rule_60min_after_min	-0.022	0.055
E2	auad	mixed	loose	p90_c2d_min	0.527	0.608
E2	auad	mixed	loose	w1_rule_60min_after_min	0.064	0.071
E2	auad	mixed	normal	p90_c2d_min	0.636	0.823

family	model	scenario	level	metric	min	max
E2	auad	mixed	normal	w1_rule_60min_after_min	0.075	0.086
E2	auad	mixed	tight	p90_c2d_min	-4.128	1.488
E2	auad	mixed	tight	w1_rule_60min_after_min	0.037	0.060
E2	auad	three_clusters	loose	p90_c2d_min	0.360	0.401
E2	auad	three_clusters	loose	w1_rule_60min_after_min	-0.176	-0.149
E2	auad	three_clusters	normal	p90_c2d_min	-0.108	0.274
E2	auad	three_clusters	normal	w1_rule_60min_after_min	-0.193	-0.171
E2	auad	three_clusters	tight	p90_c2d_min	-6.303	1.075
E2	auad	three_clusters	tight	w1_rule_60min_after_min	-0.206	0.240
E2	jusup	big_cluster	loose	p90_c2d_min	-3.154	-2.854
E2	jusup	big_cluster	loose	w1_rule_60min_after_min	-2.065	-1.816
E2	jusup	big_cluster	normal	p90_c2d_min	-2.458	-2.076
E2	jusup	big_cluster	normal	w1_rule_60min_after_min	-0.810	-0.450
E2	jusup	big_cluster	tight	p90_c2d_min	-1.826	-1.461
E2	jusup	big_cluster	tight	w1_rule_60min_after_min	0.344	0.546
E2	jusup	meetup	normal	p90_c2d_min	0.033	0.065
E2	jusup	meetup	normal	w1_rule_60min_after_min	2.272	2.455
E2	jusup	mixed	loose	p90_c2d_min	-0.708	-0.612
E2	jusup	mixed	loose	w1_rule_60min_after_min	-2.944	-2.609
E2	jusup	mixed	normal	p90_c2d_min	-1.585	-1.471
E2	jusup	mixed	normal	w1_rule_60min_after_min	-3.002	-2.683
E2	jusup	mixed	tight	p90_c2d_min	-2.485	-1.739
E2	jusup	mixed	tight	w1_rule_60min_after_min	-1.849	-1.502
E2	jusup	three_clusters	loose	p90_c2d_min	-2.366	-2.152
E2	jusup	three_clusters	loose	w1_rule_60min_after_min	-3.261	-2.853
E2	jusup	three_clusters	normal	p90_c2d_min	-3.445	-3.249
E2	jusup	three_clusters	normal	w1_rule_60min_after_min	-2.908	-2.479
E2	jusup	three_clusters	tight	p90_c2d_min	-3.435	-2.834
E2	jusup	three_clusters	tight	w1_rule_60min_after_min	-0.974	-0.561
E2	li	big_cluster	loose	p90_c2d_min	-0.024	0.014
E2	li	big_cluster	loose	w1_rule_60min_after_min	-0.091	-0.076
E2	li	big_cluster	normal	p90_c2d_min	-0.480	-0.225
E2	li	big_cluster	normal	w1_rule_60min_after_min	-0.124	-0.116
E2	li	big_cluster	tight	p90_c2d_min	-10.254	-9.179
E2	li	big_cluster	tight	w1_rule_60min_after_min	-0.075	-0.059
E2	li	meetup	normal	p90_c2d_min	0.448	0.479
E2	li	meetup	normal	w1_rule_60min_after_min	-0.549	-0.409
E2	li	mixed	loose	p90_c2d_min	-0.078	-0.048
E2	li	mixed	loose	w1_rule_60min_after_min	-0.526	-0.463
E2	li	mixed	normal	p90_c2d_min	-0.568	-0.435
E2	li	mixed	normal	w1_rule_60min_after_min	-0.566	-0.527
E2	li	mixed	tight	p90_c2d_min	-9.080	-8.788
E2	li	mixed	tight	w1_rule_60min_after_min	-0.423	-0.387
E2	li	three_clusters	loose	p90_c2d_min	-0.403	-0.294
E2	li	three_clusters	loose	w1_rule_60min_after_min	-0.226	-0.210
E2	li	three_clusters	normal	p90_c2d_min	-0.914	-0.650
E2	li	three_clusters	normal	w1_rule_60min_after_min	-0.224	-0.208
E2	li	three_clusters	tight	p90_c2d_min	-10.323	-9.793
E2	li	three_clusters	tight	w1_rule_60min_after_min	-0.063	0.016
E2	xu	big_cluster	loose	p90_c2d_min	12.024	14.199
E2	xu	big_cluster	loose	w1_rule_60min_after_min	0.216	0.266
E2	xu	big_cluster	normal	p90_c2d_min	12.018	14.758
E2	xu	big_cluster	normal	w1_rule_60min_after_min	0.884	0.922
E2	xu	big_cluster	tight	p90_c2d_min	1.812	3.972
E2	xu	big_cluster	tight	w1_rule_60min_after_min	3.303	3.395
E2	xu	meetup	normal	p90_c2d_min	3.790	3.977
E2	xu	meetup	normal	w1_rule_60min_after_min	-3.950	-1.119
E2	xu	mixed	loose	p90_c2d_min	2.337	2.495
E2	xu	mixed	loose	w1_rule_60min_after_min	-1.925	-1.676
E2	xu	mixed	normal	p90_c2d_min	0.641	0.765
E2	xu	mixed	normal	w1_rule_60min_after_min	-2.563	-2.266
E2	xu	mixed	tight	p90_c2d_min	-10.973	-10.738
E2	xu	mixed	tight	w1_rule_60min_after_min	-2.290	-1.971
E2	xu	three_clusters	loose	p90_c2d_min	3.477	3.782
E2	xu	three_clusters	loose	w1_rule_60min_after_min	-1.173	-0.915
E2	xu	three_clusters	normal	p90_c2d_min	1.298	1.802
E2	xu	three_clusters	normal	w1_rule_60min_after_min	-1.561	-1.215
E2	xu	three_clusters	tight	p90_c2d_min	-10.248	-9.559

family	model	scenario	level	metric	min	max
E2	xu	three_clusters	tight	w1_rule_60min_after_min	-0.343	0.193

W1 over 20:00 to 20:59 by return probability (clustered starts)

behavior	cell	model	difference [95% CI]
p0	big_cluster, tight	zhu	-0.04 [-0.08, -0.00]
p0	big_cluster, tight	jusup	+0.33 [+0.29, +0.37]
p0	big_cluster, tight	xu	+1.85 [+1.75, +1.96]
p0	big_cluster, normal	zhu	-0.38 [-0.41, -0.35]
p0	big_cluster, normal	jusup	-0.62 [-0.68, -0.57]
p0	big_cluster, normal	xu	+0.24 [+0.15, +0.33]
p0	big_cluster, loose	zhu	-0.29 [-0.31, -0.27]
p0	big_cluster, loose	jusup	+3.61 [+3.56, +3.67]
p0	big_cluster, loose	xu	+1.95 [+1.91, +1.99]
p0	three_clusters, tight	zhu	-0.08 [-0.12, -0.04]
p0	three_clusters, tight	jusup	+0.34 [+0.30, +0.38]
p0	three_clusters, tight	xu	-0.86 [-0.92, -0.79]
p0	three_clusters, normal	zhu	-0.05 [-0.08, -0.02]
p0	three_clusters, normal	jusup	+3.10 [+3.04, +3.16]
p0	three_clusters, normal	xu	+1.15 [+1.07, +1.23]
p0	three_clusters, loose	zhu	-0.42 [-0.45, -0.40]
p0	three_clusters, loose	jusup	+0.32 [+0.25, +0.39]
p0	three_clusters, loose	xu	-0.43 [-0.49, -0.37]
p0	mixed, tight	zhu	-0.05 [-0.09, -0.01]
p0	mixed, tight	jusup	+0.42 [+0.39, +0.46]
p0	mixed, tight	xu	-0.54 [-0.62, -0.47]
p0	mixed, normal	zhu	-0.14 [-0.16, -0.11]
p0	mixed, normal	jusup	+1.78 [+1.66, +1.89]
p0	mixed, normal	xu	+0.41 [+0.28, +0.55]
p0	mixed, loose	zhu	-0.40 [-0.43, -0.38]
p0	mixed, loose	jusup	+0.81 [+0.73, +0.90]
p0	mixed, loose	xu	+1.01 [+0.93, +1.09]
fitted	big_cluster, tight	zhu	-0.00 [-0.07, +0.06]
fitted	big_cluster, tight	jusup	+0.37 [+0.31, +0.43]
fitted	big_cluster, tight	xu	+1.75 [+1.68, +1.82]
fitted	big_cluster, normal	zhu	-5.19 [-5.28, -5.10]
fitted	big_cluster, normal	jusup	-5.44 [-5.53, -5.34]
fitted	big_cluster, normal	xu	-4.55 [-4.61, -4.50]
fitted	big_cluster, loose	zhu	-11.86 [-11.90, -11.81]
fitted	big_cluster, loose	jusup	-7.96 [-8.02, -7.89]
fitted	big_cluster, loose	xu	-3.87 [-3.92, -3.83]
fitted	three_clusters, tight	zhu	+0.29 [+0.25, +0.34]
fitted	three_clusters, tight	jusup	+0.72 [+0.67, +0.76]
fitted	three_clusters, tight	xu	+0.54 [+0.47, +0.61]
fitted	three_clusters, normal	zhu	-3.67 [-3.74, -3.60]
fitted	three_clusters, normal	jusup	-0.52 [-0.59, -0.44]
fitted	three_clusters, normal	xu	+0.18 [+0.11, +0.26]
fitted	three_clusters, loose	zhu	-3.50 [-3.55, -3.46]
fitted	three_clusters, loose	jusup	-2.76 [-2.81, -2.71]
fitted	three_clusters, loose	xu	-1.63 [-1.67, -1.60]
fitted	mixed, tight	zhu	+0.52 [+0.47, +0.58]
fitted	mixed, tight	jusup	+1.00 [+0.94, +1.06]
fitted	mixed, tight	xu	+0.92 [+0.86, +0.97]
fitted	mixed, normal	zhu	-2.74 [-2.87, -2.61]
fitted	mixed, normal	jusup	-0.82 [-0.90, -0.75]
fitted	mixed, normal	xu	+0.68 [+0.64, +0.73]
fitted	mixed, loose	zhu	-3.95 [-4.00, -3.90]
fitted	mixed, loose	jusup	-2.74 [-2.80, -2.68]
fitted	mixed, loose	xu	-0.31 [-0.35, -0.27]
p1	big_cluster, tight	zhu	-0.39 [-0.47, -0.32]
p1	big_cluster, tight	jusup	-0.02 [-0.09, +0.04]
p1	big_cluster, tight	xu	+3.22 [+3.17, +3.28]
p1	big_cluster, normal	zhu	-6.63 [-6.71, -6.55]
p1	big_cluster, normal	jusup	-6.88 [-6.96, -6.79]
p1	big_cluster, normal	xu	-2.52 [-2.56, -2.49]
p1	big_cluster, loose	zhu	-12.90 [-12.94, -12.85]
p1	big_cluster, loose	jusup	-9.00 [-9.06, -8.93]
p1	big_cluster, loose	xu	-2.01 [-2.03, -1.98]

behavior	cell	model	difference [95% CI]
p1	three_clusters, tight	zhu	+0.12 [+0.08, +0.17]
p1	three_clusters, tight	jusup	+0.55 [+0.50, +0.59]
p1	three_clusters, tight	xu	+2.17 [+2.09, +2.24]
p1	three_clusters, normal	zhu	-4.80 [-4.86, -4.74]
p1	three_clusters, normal	jusup	-1.65 [-1.71, -1.58]
p1	three_clusters, normal	xu	+0.66 [+0.59, +0.72]
p1	three_clusters, loose	zhu	-3.92 [-3.97, -3.88]
p1	three_clusters, loose	jusup	-3.18 [-3.23, -3.13]
p1	three_clusters, loose	xu	-1.13 [-1.16, -1.10]
p1	mixed, tight	zhu	+0.14 [+0.09, +0.20]
p1	mixed, tight	jusup	+0.62 [+0.56, +0.68]
p1	mixed, tight	xu	+2.31 [+2.25, +2.37]
p1	mixed, normal	zhu	-4.47 [-4.60, -4.35]
p1	mixed, normal	jusup	-2.56 [-2.63, -2.49]
p1	mixed, normal	xu	+0.24 [+0.20, +0.28]
p1	mixed, loose	zhu	-4.81 [-4.88, -4.75]
p1	mixed, loose	jusup	-3.60 [-3.66, -3.54]
p1	mixed, loose	xu	-0.37 [-0.40, -0.33]

Nearest courier's own means: `c5_evening_by_return_nearest.csv`.

C6, homes where customers live (exploratory, declared after E6 was seen)

C6. Homes where customers live (exploratory, no test; declared after the E6 results were seen)

Setting `resid`: as `indep`, except that each courier's home cell is drawn in proportion to the expected drop-offs over 17:00 to 21:00 (the customer side of the demand model, which follows resident population), from its own stream keyed by (seed, 42, courier). The 9 clustered cells, ten models, arm C at 100%, seeds 1001 to 1030. Beside E6 (`indep`) and the fitted setting at the same seeds, each read with the E6 tables and Holm within its own 144 contrasts as a reading device. Differences are model minus nearest courier, in minutes; lower is better.

setting	contrasts	established
resid	144	135
indep	144	139
fitted	144	123

Classes (cells of 72):

class	resid	indep	fitted
does not undo	72	72	65
undoes at no delivery cost	0	0	6
undoes at a delivery cost	0	0	1

Cell-model pairs that undo under any of the three settings:

cell	model	W1 60 min resid	class resid	W1 60 min indep	class indep	W1 60 min fitted	class fitted
mixed, tight	xu	-0.98	does not undo	-1.23	does not undo	-2.08	undoes at no delivery cost
mixed, normal	xu	-0.94	does not undo	-1.08	does not undo	-2.40	undoes at no delivery cost
mixed, loose	xu	-0.84	does not undo	-0.91	does not undo	-1.79	undoes at a delivery cost
three_clusters, normal	jusup	+1.61	does not undo	+1.72	does not undo	-2.65	undoes at no delivery cost
three_clusters, loose	jusup	+1.18	does not undo	+1.36	does not undo	-2.98	undoes at no delivery cost
mixed, normal	jusup	+2.94	does not undo	+2.58	does not undo	-2.76	undoes at no delivery cost
mixed, loose	jusup	+2.26	does not undo	+1.93	does not undo	-2.80	undoes at no delivery cost

Nearest courier's own W1 over the first 60 min (min, mean over the seeds)

scenario	level	resid	indep	fitted
big_cluster	tight	16.89	17.06	18.51
big_cluster	normal	17.78	18.28	21.41
big_cluster	loose	17.09	17.55	22.27
three_clusters	tight	16.41	16.34	19.50
three_clusters	normal	16.20	16.08	20.46
three_clusters	loose	15.23	15.05	19.40
mixed	tight	11.19	11.60	16.12
mixed	normal	10.76	11.12	16.46
mixed	loose	10.33	10.66	15.38

Time to undo and the second-hour check (pairs that undo under any setting, and nearest courier)

The time to undo is measured against the same model's scattered start under the fitted setting, for all three settings (as in E6); the second-hour check applies to the cells classed as undoing.

setting	scenario	level	model	class	time_to_undo_median	censored_count	nearest_time_to_undo_median	second_hour_check
fitted	big_cluster	loose	nearest	nearest courier	240.000	30	240.000	

setting	scenario	level	model	class	time_to_undo_median	censored_count	nearest_time_to_undo_median	second_hour_check
indep	big_cluster	loose	nearest	nearest courier	136.000	0	136.000	
resid	big_cluster	loose	nearest	nearest courier	112.000	0	112.000	
fitted	big_cluster	normal	nearest	nearest courier	74.000	0	74.000	
indep	big_cluster	normal	nearest	nearest courier	93.000	0	93.000	
resid	big_cluster	normal	nearest	nearest courier	90.000	0	90.000	
fitted	big_cluster	tight	nearest	nearest courier	54.000	0	54.000	
indep	big_cluster	tight	nearest	nearest courier	55.500	0	55.500	
resid	big_cluster	tight	nearest	nearest courier	56.000	0	56.000	
fitted	mixed	loose	nearest	nearest courier	240.000	30	240.000	
indep	mixed	loose	nearest	nearest courier	74.000	0	74.000	
resid	mixed	loose	nearest	nearest courier	68.000	0	68.000	
fitted	mixed	normal	nearest	nearest courier	113.000	0	113.000	
indep	mixed	normal	nearest	nearest courier	67.500	0	67.500	
resid	mixed	normal	nearest	nearest courier	63.000	0	63.000	
fitted	mixed	tight	nearest	nearest courier	64.000	0	64.000	
indep	mixed	tight	nearest	nearest courier	61.500	0	61.500	
resid	mixed	tight	nearest	nearest courier	57.500	0	57.500	
fitted	three_clusters	loose	nearest	nearest courier	240.000	30	240.000	
indep	three_clusters	loose	nearest	nearest courier	171.500	5	171.500	
resid	three_clusters	loose	nearest	nearest courier	240.000	30	240.000	
fitted	three_clusters	normal	nearest	nearest courier	240.000	17	240.000	
indep	three_clusters	normal	nearest	nearest courier	137.000	0	137.000	
resid	three_clusters	normal	nearest	nearest courier	146.500	10	146.500	
fitted	three_clusters	tight	nearest	nearest courier	60.000	0	60.000	
indep	three_clusters	tight	nearest	nearest courier	64.000	0	64.000	
resid	three_clusters	tight	nearest	nearest courier	64.000	0	64.000	
fitted	mixed	loose	jusup	undoes at no delivery cost	240.000	23	240.000	holds after the first hour
indep	mixed	loose	jusup	does not undo	240.000	23	74.000	
resid	mixed	loose	jusup	does not undo	240.000	23	68.000	
fitted	mixed	normal	jusup	undoes at no delivery cost	126.000	0	113.000	undoes in the first hour, not after
indep	mixed	normal	jusup	does not undo	126.000	0	67.500	
resid	mixed	normal	jusup	does not undo	126.000	0	63.000	
fitted	three_clusters	loose	jusup	undoes at no delivery cost	240.000	30	240.000	holds after the first hour
indep	three_clusters	loose	jusup	does not undo	240.000	30	171.500	
resid	three_clusters	loose	jusup	does not undo	240.000	30	240.000	
fitted	three_clusters	normal	jusup	undoes at no delivery cost	240.000	29	240.000	undoes in the first hour, not after
indep	three_clusters	normal	jusup	does not undo	240.000	29	137.000	
resid	three_clusters	normal	jusup	does not undo	240.000	29	146.500	
fitted	mixed	loose	xu	undoes at a delivery cost	240.000	30	240.000	holds after the first hour
indep	mixed	loose	xu	does not undo	60.000	0	74.000	
resid	mixed	loose	xu	does not undo	55.500	0	68.000	
fitted	mixed	normal	xu	undoes at no delivery cost	240.000	30	113.000	undoes in the first hour, not after
indep	mixed	normal	xu	does not undo	56.000	0	67.500	
resid	mixed	normal	xu	does not undo	53.000	0	63.000	
fitted	mixed	tight	xu	undoes at no delivery cost	86.000	0	64.000	undoes in the first hour, not after
indep	mixed	tight	xu	does not undo	53.500	0	61.500	
resid	mixed	tight	xu	does not undo	52.000	0	57.500	

W1 over 20:00 to 20:59, model minus nearest courier, mean of the nine cells

model	resid	indep	fitted
auad	-0.020	0.070	0.290
backpressure	-0.160	-0.230	-0.480
foodmatch	0.000	-0.010	1.240
forecast	-0.570	0.080	0.150
jahanshahi	-0.050	-0.080	-0.060
jusup	3.570	4.190	-2.050
li	-0.530	-0.700	0.070
xu	-0.550	0.120	-0.710
zhu	2.240	2.860	-3.370

The DQN row is descriptive.

Every number per cell: `c6_resid_contrasts.csv`, `c6_resid_classes.csv`, `c6_resid_undo.csv`, `c6_resid_evening.csv`, `c6_resid_nearest.csv`.

C7, the fitted behavior's bands on a wider grid

C7. The fitted behavior's bands on a wider grid (Paper A, Table 1)

Bands are 95% bootstrap intervals over the Meituan moments (the earlier fit's procedure: moments, seeds, objective, bootstrap picks over every grid point run). The wide grid adds two steps beyond the lowest and the highest value the earlier fit explored for every fitted parameter, then grows on any side a band end still touches (4 rounds; at most 8).

parameter	earlier band	wide-grid band	wide grid explored	end still touches the grid	point: earlier, wide
return probability	[0.675, 0.825]	[0.675, 0.825]	[0.45, 0.975]	no	0.75, 0.75
decision delay (s)	[29.4, 42]	[29.4, 42]	[15, 117.6]	no	42, 42
cruise radius (m)	[2000, 2800]	[1979, 2800]	[714.286, 7056]	no	2000, 2800
stay constant (beta0)	[-3.5, -2]	[-6, -2.5]	[-6.5, -1]	no	-3, -3.5

Band ends that still touch the wide grid: none. Band ends one value from the edge of the explored values: stay constant (beta0) (low). Values added in the rounds were tried only across the other parameters' values inside their bands, so a band end there may be too narrow: stay constant (beta0): lowest value at 18 points against a median of 512 per value. Objective values are not comparable across fits: the moment weights depend on the grid's coarse points. Point estimate: moved for cruise radius (m), stay constant (beta0). As instructed, nothing in the frozen behavior changes: every run keeps the frozen values. 0 of 156 grid points shared with the earlier fit reproduce its moments exactly.

The Ashkrof variant's bands on a wider grid

The Ashkrof variant's bands on a wider grid

Bands are 95% bootstrap intervals over the Meituan moments (the earlier fit's procedure: moments, seeds, objective, bootstrap picks over every grid point run). The wide grid adds two steps beyond the lowest and the highest value the earlier fit explored for every fitted parameter, then grows on any side a band end still touches (4 rounds; at most 8).

parameter	earlier band	wide-grid band	wide grid explored	end still touches the grid	point: earlier, wide
return probability	[0.65, 0.85]	[0.675, 0.85]	[0.4, 1]	no	0.75, 0.75
decision delay (s)	[42, 42]	[42, 42]	[42, 42]	fixed	42, 42
cruise radius (m)	[2000, 2000]	[2000, 2000]	[2000, 2000]	fixed	2000, 2000
stay constant (beta0)	[-4, -2]	[-6.5, -2.7375]	[-7, -1]	no	-3, -4.5

Band ends that still touch the wide grid: none. Band ends one value from the edge of the explored values: stay constant (beta0) (low). Objective values are not comparable across fits: the moment weights depend on the grid's coarse points. Point estimate: moved for stay constant (beta0). Part 3 ran again at the wide-grid values (setting `ashkrofw`, files `c3w_ashkrof_*`); the first runs stay beside them. 25 of 25 grid points shared with the earlier fit reproduce its moments exactly.

Part 3 again, at the wide-grid recalibration

Part 3 again, at the wide-grid recalibration (exploratory, no test)

The first recalibration's grid cut the beta0 interval. The wide-grid recalibration (`ashkrof_fit_wide/`, setting `ashkrofw`) moved beta0 from -3 to -4.5, with the band [-6.5, -2.7375]; the first runs at beta0 -3 lie inside it and stay beside these (`c3_ashkrof_*`). The objective is flat in beta0, so the two points fit the moments about equally well. Part 3 ran again at the new values; everything else as in part 3. The fitted columns `fitted_wide_ci_*` are the fitted behavior's bands on its own wide grid (C7).

Recalibrated values beside the fitted behavior (95% bootstrap intervals over the Meituan moments):

parameter	fitted	fitted_ci_lo	fitted_ci_hi	fitted_wide_ci_lo	fitted_wide_ci_hi	ashkrof	ashkrof_ci_lo	ashkrof_ci_hi	ashkrof_note
p_return	0.750	0.675	0.825	0.675	0.825	0.750	0.675	0.850	recalibrated
asc_stay	-3.000	-3.500	-2.000	-6.000	-2.500	-4.500	-6.500	-2.738	recalibrated
epoch_s	42.000	29.400	42.000	29.400	42.000	42.000	42.000	42.000	fixed at the fitted value
cruise_radius_m	2000.000	2000.000	2800.000	1979.000	2800.000	2000.000	2000.000	2000.000	fixed at the fitted value
part_time_stay	nan	nan	nan	nan	nan	0.000	nan	nan	set to 0 (Scenario 1 Primary otherwise)
objective	2316.829	nan	nan	nan	nan	2484.122	nan	nan	

Moments (Meituan peak hours, the fitted behavior and the Ashkrof setting):

moment	in_fit	meituan_peak	meituan_all_day	simulated_fitted	simulated_ashkrof
exc_1000_15-30	False	0.431	0.443	0.494	0.497
exc_1000_5-15	False	0.335	0.345	0.129	0.131
exc_1000_<5	False	0.087	0.088	0.010	0.009
exc_1000_>=30	False	0.391	0.448	0.594	0.588
exc_1000_all	False	0.285	0.323	0.200	0.199
exc_2000_15-30	True	0.514	0.521	0.606	0.607
exc_2000_5-15	True	0.420	0.427	0.269	0.274
exc_2000_<5	True	0.098	0.105	0.049	0.048
exc_2000_>=30	True	0.490	0.539	0.695	0.690
exc_2000_all	False	0.369	0.406	0.308	0.307
exc_3000_15-30	False	0.501	0.522	0.587	0.584
exc_3000_5-15	False	0.399	0.414	0.242	0.251
exc_3000_<5	False	0.086	0.088	0.051	0.052
exc_3000_>=30	False	0.528	0.559	0.687	0.689
exc_3000_all	False	0.386	0.424	0.312	0.313

moment	in_fit	meituan_peak	meituan_all_day	simulated_fitted	simulated_ashkrof
move_mean	nan	nan	nan	nan	1169.973
move_q10	True	43.393	73.282	0.000	0.000
move_q25	True	312.686	427.585	263.037	287.888
move_q50	True	910.804	1070.989	678.874	692.966
move_q75	True	1751.876	1961.133	1515.168	1522.629
move_q90	True	2748.672	2959.550	2864.637	2862.038
ret_1000_15-30	False	0.532	0.551	0.595	0.598
ret_1000_5-15	False	0.433	0.448	0.212	0.217
ret_1000_<5	False	0.130	0.131	0.035	0.034
ret_1000_>=30	False	0.484	0.555	0.687	0.681
ret_1000_all	False	0.366	0.413	0.263	0.263
ret_2000_15-30	True	0.690	0.705	0.788	0.787
ret_2000_5-15	True	0.600	0.611	0.402	0.411
ret_2000_<5	True	0.162	0.171	0.085	0.086
ret_2000_>=30	True	0.653	0.719	0.850	0.845
ret_2000_all	False	0.516	0.564	0.416	0.416
ret_3000_15-30	False	0.711	0.736	0.798	0.794
ret_3000_5-15	False	0.606	0.627	0.384	0.400
ret_3000_<5	False	0.158	0.160	0.093	0.095
ret_3000_>=30	False	0.728	0.768	0.876	0.876
ret_3000_all	False	0.563	0.610	0.438	0.440
stay_15-30	True	0.035	0.040	0.038	0.026
stay_5-15	True	0.093	0.082	0.069	0.053
stay_<5	True	0.628	0.612	0.563	0.553
stay_>=30	True	0.030	0.031	0.041	0.028

Contrasts and classes

All ten models ran, seeds 1001 to 1030; the eight tested models are read against nearest courier, beside the fitted behavior's full-run rows at the same seeds, read the same way. Smallest effects per behavior: 10% of nearest courier's mean under that behavior (the fitted pilot for the fitted rows; seeds 1001 to 1030 of this part for `ashkrofw`, which had no pilot). Holm within the E1-shaped and E2-shaped sets of each behavior, as a reading device. The DQN is a descriptive row (mean differences, no interval). Contrasts: `c3w_ashkrof_contrasts.csv`; classes: `c3w_ashkrof_classes.csv`; DQN: `c3w_ashkrof_dqn.csv`.

W1 is W1 over the evening on Q1 cells and W1 over 60 min on Q2 cells; differences are model minus nearest courier, in minutes; lower is better.

input	holm_set	contrasts	established
ashkrofw	E1-shaped	96	85
ashkrofw	E2-shaped	160	134
fitted	E1-shaped	96	85
fitted	E2-shaped	160	132

Classes by model (cells):

model	level_1	ashkrofw	fitted
auad	does not undo	10	10
auad	does not prevent	6	6
backpressure	does not undo	10	10
backpressure	does not prevent	4	4
backpressure	prevents at no delivery cost	2	2
foodmatch	does not undo	10	10
foodmatch	does not prevent	6	6
forecast	does not undo	10	10
forecast	prevents at no delivery cost	4	4
forecast	does not prevent	2	2
jusup	does not prevent	6	6
jusup	does not undo	6	6
jusup	undoes at no delivery cost	4	4
li	does not undo	10	10
li	prevents at no delivery cost	4	4
li	does not prevent	2	2
xu	does not undo	6	6
xu	does not prevent	4	4
xu	undoes at no delivery cost	2	2
xu	undoes at a delivery cost	2	2
xu	prevents at a delivery cost	1	1
xu	prevents at no delivery cost	1	1
zhu	does not undo	10	10
zhu	does not prevent	6	6

Classes that change: 0 of 128.

None.

Files

One CSV per table, in this folder:

- `c1_e6_classes.csv`
- `c1_e6_contrasts.csv`
- `c1_e6_dqn.csv`
- `c1_e6_evening.csv`
- `c1_e6_undo.csv`
- `c2_stay.csv`
- `c3_ashkrof_bands.csv`
- `c3_ashkrof_classes.csv`
- `c3_ashkrof_contrasts.csv`
- `c3_ashkrof_dqn.csv`
- `c3_ashkrof_fit.csv`
- `c3_ashkrof_moments.csv`
- `c3_ashkrof_nearest_means.csv`
- `c3w_ashkrof_classes.csv`
- `c3w_ashkrof_contrasts.csv`
- `c3w_ashkrof_dqn.csv`
- `c3w_ashkrof_fit.csv`
- `c3w_ashkrof_moments.csv`
- `c3w_ashkrof_nearest_means.csv`
- `c4_noise_classes.csv`
- `c4_noise_contrasts.csv`
- `c4_noise_identity.csv`
- `c5_evening_by_return.csv`
- `c5_evening_by_return_nearest.csv`
- `c5_evening_truncated.csv`
- `c5_per_policy.csv`
- `c5_w1_windows.csv`
- `c6_resid_classes.csv`
- `c6_resid_contrasts.csv`
- `c6_resid_evening.csv`
- `c6_resid_nearest.csv`
- `c6_resid_undo.csv`
- `c7_bands.csv`

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